HIGH-LEVEL DESIGN DOCUMENT

LeadSpark CRM

Business Architecture & System Design

> A business-centric overview of the LeadSpark CRM platform — covering its key capability domains, business decision-making logic, integration ecosystem, and communication architecture. This document is intended for product, business, and engineering leadership.

DOCUMENT TYPE	PRODUCT	AUDIENCE	DATE
High-Level Design (HLD)	LeadSpark CRM	Business & Engineering Leaders	April 2026

HLD

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Infrastructure

Infrastructure, environments, and release governance

SECTION 01 — CONTEXT

What is LeadSpark CRM?

The business problem this platform solves and the value it delivers to sales teams.

Purpose: LeadSpark CRM is a purpose-built Sales Development Representative (SDR) workflow platform. It takes raw prospect leads — from Salesforce, CSV uploads, or manual entry — and guides each SDR through a structured, AI-assisted research-to-calling pipeline until a meeting is scheduled or the lead is disqualified.

🎯

Problem Being Solved

SDR teams lack a structured, data-informed workflow. Leads fall through the cracks, calling begins without research, and there's no unified view of pipeline health.

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How LeadSpark Solves It

Forces research before calling, auto-generates AI intelligence on each prospect, enforces a gated pipeline, and gives leadership real-time visibility into every SDR's performance.

📈

Business Outcome

Higher meeting conversion rates. Consistent SDR behavior. Full auditability of every interaction. Seamless sync with Salesforce as the system of record.

SECTION 02 — PEOPLE

Users & Roles

LeadSpark has three user personas with distinct responsibilities and access boundaries. Roles are enforced at every point in the system — not just in the UI.

🔴

Super Admin

The platform owner. Manages the entire system including:

- Uploading and deleting leads
- Managing all users and access
- Configuring Salesforce, Aircall, Nylas
- Viewing all pods, all leads, all logs
- Triggering Salesforce sync
- Promoting/demoting user roles

Typically: *Sales Ops / VP of Sales*

🟡

Pod Admin

The team lead. Manages a subset of SDRs organized into a "Pod":

- Assigns leads to SDRs in their pod
- Moves leads backward in the pipeline
- Disqualifies leads
- Views pod-level performance metrics
- Cannot manage other pods or system settings

Typically: *SDR Manager / Team Lead*

🟢

SDR

The operator. Works their assigned lead queue:

- Runs AI research on each lead
- Calls leads via Aircall or manual dial
- Logs call outcomes
- Emails leads via connected inbox
- Adds notes, tasks, and attachments
- Cannot move leads backward

Typically: *Sales Development Representative*

SECTION 03 — ARCHITECTURE

Key Capability Domains

The platform is organized around 7 distinct functional domains. Each domain owns a clear business responsibility and operates independently while communicating with others through defined interfaces.

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The 7 capability domains of LeadSpark CRM. Identity & Access underpins all other domains.

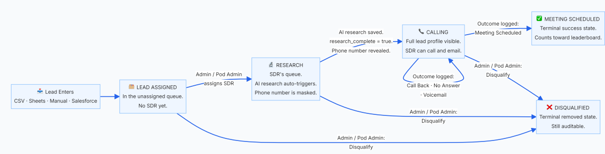
Domain	Business Responsibility	Key Business Value
🏗️ **Lead Management**	Single source of truth for all prospects. Supports CSV upload, Google Sheets import, manual creation, and Salesforce pull. Full lifecycle tracking.	Every lead is accounted for — no lost prospects, no duplicate effort.
🤖 **AI Research Engine**	Auto-triggers LLM research on each lead when an SDR opens their profile. Populates strategic talking points, pain points, and call strategy. Phone number is masked until research is complete.	SDRs enter every call prepared. Research quality replaces cold calling.
📞 **Communication Hub**	Integrates Aircall for VoIP calling and Nylas for email. All calls and email threads are logged against the lead — creating a unified interaction record.	All communication happens inside the CRM. Zero context-switching for SDRs.
🔄 **Salesforce Sync**	Bi-directional sync with Salesforce. Pulls leads in. Pushes status updates, call outcomes, and SDR	Salesforce stays current automatically.

Domain	Business Responsibility	Key Business Value
	metrics back. Every sync operation is logged with success/failure.	Leadership reports are always accurate.
👥 **Team & Pod Engine**	Organizes SDRs into named "Pods" managed by Pod Admins. Automates lead assignment via round-robin with cap enforcement. Tracks per-SDR capacity and activity.	Fair, transparent lead distribution. Pod Admins have autonomy without full system access.
📊 **Analytics & Visibility**	Real-time leaderboard, daily metrics dashboard, activity feed, audit logs, and Salesforce operation history. Exportable to CSV.	Leadership sees exactly what's happening without manual reporting.
🔒 **Identity & Access**	Google SSO for login. JWT tokens for API authentication. Role-level access control enforced server-side on every API call. Allowlist controls who may log in.	Only authorized, verified users can access the system. No unauthorized data exposure.

SECTION 04 — BUSINESS FLOW

The Lead Lifecycle

A lead enters the system and progresses through a gated pipeline. Each stage has defined entry conditions, permitted actors, and allowed transitions. The pipeline cannot be skipped.



Stage-by-Stage Business Rules

1

Lead Assigned — The Inbox Stage

A lead enters when uploaded (CSV/Google Sheets), manually created by an admin, or pulled from Salesforce. It sits in an unassigned queue until a Super Admin or Pod Admin assigns it to an SDR. Assignment is governed by the Lead Assignment Engine (see Section 5). The SDR can see the lead exists but cannot see the phone number yet.

2

Research — The Intelligence Gate

Once assigned, the SDR opens the lead profile. The AI Research Engine immediately fires a call to Groq AI, building a research brief from the lead's company, title, LinkedIn, and other profile data. The phone number remains masked. The SDR reviews and validates the AI output. Only when research is marked complete does the phone number become visible and Calling becomes accessible.

3

Calling — The Active Engagement Stage

The SDR can now call (via Aircall or standard dial), email (via connected Nylas inbox), add notes, create follow-up tasks, and upload attachments. Every call must be logged with an outcome. Certain outcomes (Call Back Later, No Answer) automatically reduce the lead's priority score so fresh leads stay visible. This stage can loop indefinitely — a lead stays in Calling until an outcome that triggers a transition occurs.

4

Meeting Scheduled — Terminal Success

When a call is logged with outcome "Meeting Scheduled," the system automatically transitions the lead to this terminal status. It appears on the SDR leaderboard, is pushed to Salesforce, and counts toward daily metrics. The SDR can report No-Shows (tracked as a count on the lead), but the status does not change — Meeting Scheduled is permanent.

5

Disqualified — Terminal Removal

Admins and Pod Admins may disqualify a lead at any pipeline stage — no phone number, wrong industry, already a customer, etc. The lead is removed from active queues but remains fully auditable in the database. Status history is preserved. This action is irreversible by SDRs — only admins can act on disqualified leads.

Key Business Rule: The pipeline is **forward-only for SDRs**. SDRs cannot move leads backward or skip stages. Only Pod Admins and Super Admins can reverse a transition — and every reversal is logged with timestamp and actor for full auditability.

SECTION 05 — INTELLIGENCE

Business Decision-Making Systems

LeadSpark contains several automated decision engines that remove manual judgment calls from routine operations. These systems enforce business rules consistently at scale.

5.1 — Lead Assignment Engine

What it decides: Which SDR in a pod receives a newly assigned lead, and when an SDR has reached capacity.

⚙️ Assignment Rules (in priority order)

- **Pod Scope:** Leads can only be assigned to SDRs within the configured pod. Cross-pod assignment is not permitted.
- **Active Lead Cap:** Each SDR has a configurable maximum number of active (non-terminal) leads. An SDR at capacity is skipped.
- **Round-Robin Distribution:** Among eligible SDRs (in pod + under cap), leads are distributed in rotation to ensure fairness.
- **Multi-Pod SDR Flag:** Optionally, an SDR may belong to multiple pods. When enabled, their cap applies across all pods combined.
- **Auto-Assign:** Admins can trigger a bulk auto-assign that distributes the entire unassigned queue across eligible SDRs in a single operation.

Scenario	System Decision
SDR is under cap and in the pod	Eligible for assignment in the round-robin rotation
SDR is at maximum active lead cap	Skipped — not assigned until existing leads move to terminal status
All SDRs in pod are at cap	Lead stays in unassigned queue — admin is responsible for resolution

Scenario	System Decision
Manual assignment by Pod Admin	Bypasses round-robin — assigned to specified SDR (cap still enforced optionally)

5.2 — Lead Priority & Scoring Engine

What it decides: The order in which an SDR sees their leads in the list view. Higher priority leads appear first.

📊 **Priority Score Rules**

- **Initial Priority:** All leads enter with the same base priority score.
- **Negative Call Outcomes:** Logging "Call Back Later" or "No Answer" automatically decrements the priority score. This pushes the lead down the SDR's queue, allowing fresher leads to surface.
- **Manual Override:** Admins can manually reprioritize a specific lead — boosting or deprioritizing it regardless of call history.
- **Priority Tiers:** Leads are visually categorized into High / Medium / Low priority tiers based on their score, with color-coded badges in the lead list.

5.3 — Pipeline Gate: Research Completion Check

What it decides: Whether an SDR is allowed to advance a lead from Research to Calling status.

🚦 **Gate Logic**

- The lead profile has a **research_complete** flag (true/false).
- When the SDR saves the AI-generated research data, this flag is set to **true**.
- Until this flag is true, the **phone number is masked** in the UI — the SDR cannot see or dial it.

- The "Move to Calling" action is blocked at the API level — not just the UI — until **research_complete = true**.
- This gate cannot be bypassed by SDRs. Admins can manually set a lead to Calling if operationally required.

5.4 — Call Outcome Classification Engine

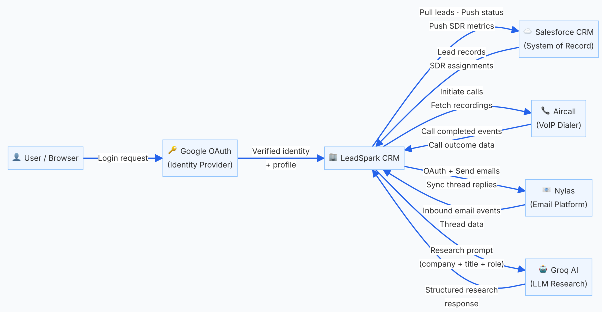
What it decides: What happens to a lead automatically based on the outcome an SDR logs after a call.

Call Outcome Logged	Automatic System Action	Status Change?
Meeting Scheduled	Lead status → Meeting Scheduled. Leaderboard count +1. Pushed to Salesforce.	✅ Yes — Terminal
Meeting Confirmed	Escalation of Meeting Scheduled. Company marked resolved.	✅ Yes — Terminal
Call Back Later	Priority score decremented. Lead moves down the queue.	❌ No change
No Answer	Priority score decremented. Lead moves down the queue.	❌ No change
Left Voicemail	Call logged for audit. No priority change.	❌ No change
Not Interested	Call logged. Admin may subsequently disqualify.	❌ No change (admin action needed)
Wrong Number	Call logged. Admin may subsequently disqualify.	❌ No change (admin action needed)

SECTION 06 — INTEGRATIONS

Integration Ecosystem

LeadSpark connects to five external platforms. Each integration has a defined data ownership boundary, communication direction, and failure handling strategy.



☁️ Salesforce CRM

System of Record · Bi-directional Sync

What LeadSpark pulls	Lead records, company data, SDR assignments
What LeadSpark pushes	Status changes, call outcomes, SDR performance metrics
Trigger	Manual trigger by Super Admin or scheduled sync
Failure Handling	Every operation logged as success or failure with payload in SF Audit Log

📞 Aircall

VoIP Dialer · Inbound Webhooks

What LeadSpark initiates	Outbound call from SDR's browser to lead's number
What Aircall sends back	Call completion event, duration, recording URL, transcript URL
Trigger	SDR clicks "Call" in lead detail page
Failure Handling	Manual call logging fallback available if Aircall is unavailable
Authentication	API Key + API ID (encrypted at rest)

Authentication	Username + Password + Security Token (encrypted at rest)

📧 **Nylas**

Email Platform · Per-user OAuth

What LeadSpark sends	Outbound emails from SDR's connected inbox to leads
What Nylas delivers	Inbound email replies (via webhook) synced to lead's email tab
Per-User Setup	Each SDR connects their own email account via OAuth — tokens stored encrypted
Failure Handling	Webhook challenge verification maintained; token refresh on expiry
Authentication	Per-user OAuth 2.0 grant + Nylas API Key (encrypted at rest)

🤖 **Groq AI (LLM)**

Research Generation · Outbound Only

What is sent	Lead's company, title, LinkedIn URL, and persona data as a prompt
What is returned	Structured research: pain points, talking points, call strategy, objection handling
Trigger	Automatically on page load when SDR opens a lead in Research status
Failure Handling	SDR can manually trigger a retry. Research fields can be edited manually.
Authentication	Groq API Key (environment variable, server-side only)

🔑 **Google OAuth 2.0**

Identity Provider · Login Only · Outbound

What it does	Verifies the user's Google identity and returns their email + name + profile picture

Access Control	**Email must be on the CRM allowlist — unallowed emails are rejected even after Google verification**
Token Issued	**On success, CRM issues its own JWT containing the user's ID, role, and email — valid for the session**
Authentication	**OAuth 2.0 client credentials (configured per environment)**

SECTION 07 — COMMUNICATION

Communication Architecture

LeadSpark uses four distinct communication patterns depending on the latency, reliability, and user-experience requirements of each interaction.

🔄 Synchronous REST (Primary)

All user-facing operations — loading lead lists, saving research, logging calls, sending emails — use standard HTTPS REST API calls. The browser waits for a response before updating the UI.

When used: Any operation where the user needs immediate confirmation.
Auth: JWT Bearer token on every request.
Latency target: < 500ms for list/read, < 1s for writes.

User actions | Lead CRUD | Call logging

📨 Inbound Webhooks (Event-Driven)

External systems push events into LeadSpark via registered webhook endpoints. LeadSpark processes these events asynchronously without user involvement.

Aircall: Sends *call.completed* and *call.answered* events when a call ends.
Nylas: Sends inbound email receipt events when a lead replies to an SDR's email.

Aircall events | Nylas email sync

⏰ Scheduled Background Jobs

Four recurring background jobs run on a schedule, completely independent of user actions. They handle operations that should be invisible to users.

Health Check: Verifies system connectivity.
Daily Summaries: Aggregates SDR metrics for the dashboard.
Log Cleanup: Removes aged log entries to control DB size.

🌐 LLM Streaming (AI Research)

AI research responses from Groq are streamed to the browser as the LLM generates them — users see content appear word-by-word rather than waiting for the full response. This is automatically triggered when an SDR opens a lead in Research status.

Trigger: Page load for a Research-status lead.
Response: Server-Sent Events or chunked transfer.

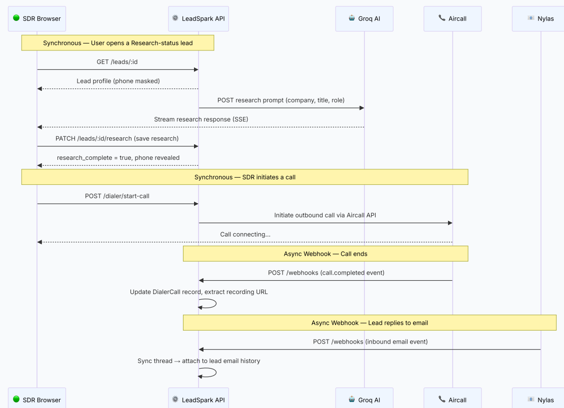
Keep-Alive Ping: Prevents the hosting platform from cold-starting.

APScheduler

Background threads

Streaming

Auto-triggered



End-to-end communication flow for a typical SDR session — research → call → email reply.

SECTION 08 — DATA

Data Ownership & Flow

Understanding which system is the authoritative source for each type of data is critical for resolving conflicts, planning integrations, and ensuring data integrity.

Data Type	Authoritative Source	How it enters LeadSpark	How it flows out
Lead Profile Data (name, company, email, phone, title)	Salesforce CRM	Salesforce sync pull, OR manual CSV/Sheets upload	Status + outcome changes pushed back to Salesforce on sync
AI Research Content (talking points, pain points, strategy)	LeadSpark CRM	Generated by Groq LLM, validated and saved by SDR	Does not leave LeadSpark (SDR use only)
Call Records (date, duration, outcome, recording)	LeadSpark CRM	SDR manual log, OR auto-created from Aircall webhook	Outcome summary pushed to Salesforce on sync
Email Threads	Nylas	Outbound: composed in LeadSpark, sent via Nylas. Inbound: Nylas webhook pushes to LeadSpark.	Does not leave LeadSpark after ingestion
SDR Performance Metrics	LeadSpark CRM	Aggregated from call logs + status events by daily summary job	Pushed to Salesforce as SDR activity data on sync
User Identities	Google	Google OAuth login verifies identity; CRM	Does not leave LeadSpark after

Data Type	Authoritative Source	How it enters LeadSpark	How it flows out
		creates/updates local user record	authentication
OAuth Tokens (Salesforce, Nylas)	LeadSpark CRM	Obtained via OAuth flow; stored encrypted in DB using Fernet symmetric encryption	Used only server-side; never exposed to browser or logs
Audit & Activity Logs	LeadSpark CRM	Auto-generated by every state change, login event, and SF operation	Exportable to CSV by Super Admin; not pushed to external systems

Data Integrity Principle: Salesforce is the system of record for lead and prospect data. LeadSpark enriches that data with AI research, interaction history, and workflow state. On sync, LeadSpark's workflow state is pushed back to Salesforce — never the reverse for AI or interaction data.

SECTION 09 — GOVERNANCE

Observability & Governance

Every significant action in LeadSpark is recorded. The system provides four layers of observability — supporting both operational troubleshooting and business compliance.

🔎

Salesforce Audit Log

Every Salesforce API operation (lead pull, status push, SDR sync) is logged with full request/response payload, success/failure status, and timestamp.

Accessible by: Super Admin only.
Use cases: Debug failed syncs; replay failed pushes; compliance review.

SF Operations | Success/Failure | Exportable CSV

📋

User Activity Log

Tracks all meaningful SDR and admin actions: calls logged, emails sent, leads moved, research completed, status changes, note additions.

Accessible by: Super Admin (all users), Pod Admin (own pod).
Use cases: Performance management; dispute resolution.

SDR Actions | Status Changes | Filterable

🔐

Login Log

Records every login event with user email, IP address, user agent, and timestamp. Can identify unauthorized access attempts or audit access from unexpected locations.

Accessible by: Super Admin only.

Security | Access Tracking

📊

Lead Status History

Every time a lead's status changes (forward or backward), it is recorded with the previous status, new status, who made the change, and when. Creates a full lifecycle audit trail per lead.

Accessible by: All roles (for their accessible leads).

Per Lead

Actor Tracked

Immutable

Deployment & Environments

LeadSpark runs on two isolated cloud environments. All changes flow through Staging before reaching Production. No direct production deployments are permitted.

🟡

Staging Environment

URL: conversive-crm-staging.onrender.com
Branch: develop
Purpose: All development, testing, bug fixes, and feature validation happen here. SDRs and admins are never expected to use this environment. It auto-deploys when code is pushed to the *develop* branch.

Auto-deploy on develop push

🔴

Production Environment

URL: conversive-crm-prod.onrender.com
Branch: main
Purpose: Live system used by real SDRs. Auto-deploys when code is merged to *main* via a Pull Request. A PR to main requires explicit approval — it is never done automatically.

Explicit approval required

Release Governance: Before any feature reaches Production, it must:

- Be fully tested on Staging (functional + edge cases)
- Have User Guide documentation updated in `docs/user_guide.md`
- Have Release Notes updated in `docs/release-notes.md`
- Have in-app release notes updated for SDRs to see on login
- Receive explicit approval from the platform owner before the PR to main is merged